```python
# Implementation of DBSCAN Clustering for Cluster Identification and
Outlier Detection

import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import DBSCAN
from sklearn.metrics import silhouette_score
import numpy as np


print("SHRAWANI KHANDAGALE")
print("Roll No.-69\n")

# 1. Generate Dataset
X, _ = make_blobs(n_samples=200, centers=3, cluster_std=0.6,
random_state=42)

# Add some outliers
outliers = np.array([[8, 10], [-8, -10], [10, -8]])
X = np.vstack((X, outliers))

# 2. Standardize the Data
X_scaled = StandardScaler().fit_transform(X)

# 3. Apply DBSCAN Clustering
model = DBSCAN(eps=0.3, min_samples=5)
labels = model.fit_predict(X_scaled)

# 4. Identify Clusters and Outliers
clusters = set(labels) - {-1}
outlier_count = list(labels).count(-1)

print("Number of clusters:", len(clusters))
print("Number of outliers:", outlier_count)
print("Cluster labels:", labels)

# 5. Visualize Clusters and Outliers
plt.figure(figsize=(8, 5))

colors = ['blue', 'green', 'orange', 'purple', 'brown']

for label in sorted(set(labels)):
    points = X_scaled[labels == label]

    if label == -1:
        plt.scatter(points[:, 0], points[:, 1],
                    color='red', marker='x', s=100, label='Outliers')
    else:
        plt.scatter(points[:, 0], points[:, 1],
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                    s=50, label=f'Cluster {label}')

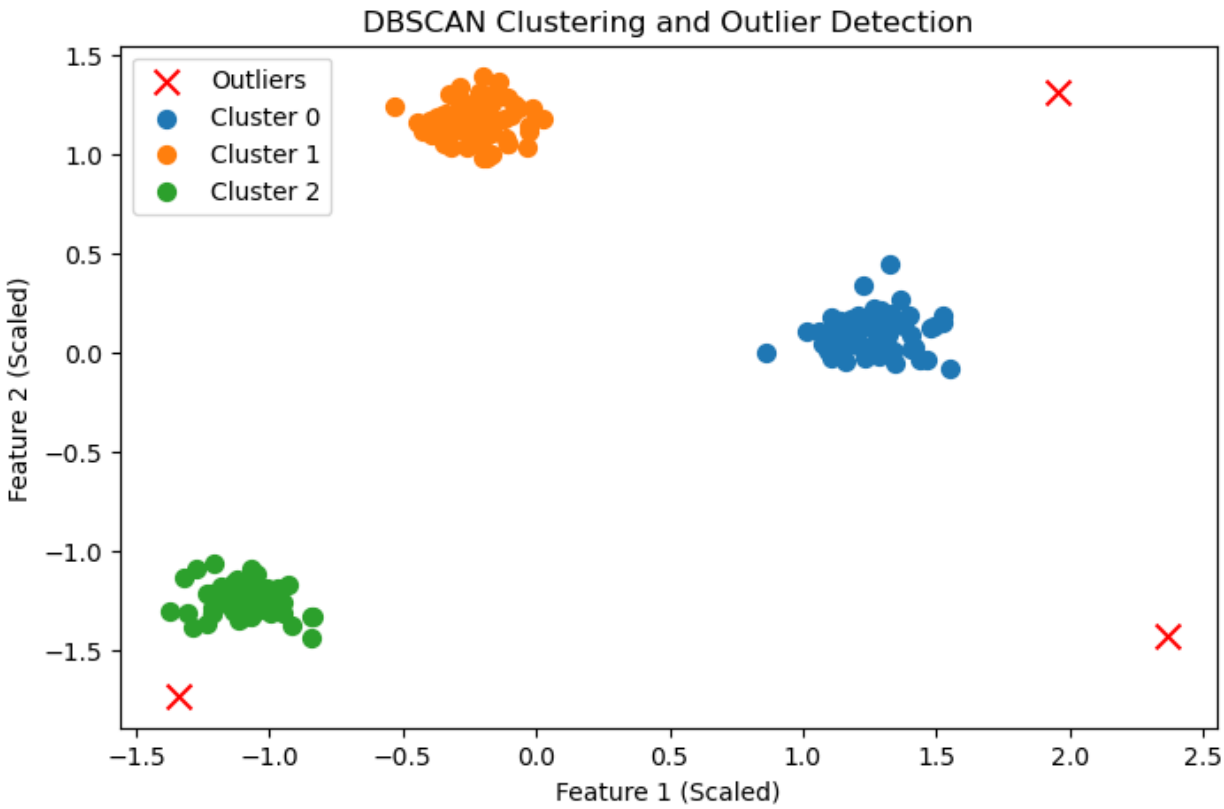
plt.title("DBSCAN Clustering and Outlier Detection")
plt.xlabel("Feature 1 (Scaled)")
plt.ylabel("Feature 2 (Scaled)")
plt.legend()
plt.show()

# 6. Calculate Silhouette Score
if len(clusters) > 1 and len(clusters) < len(X_scaled):
    score = silhouette_score(X_scaled[labels != -1], labels[labels != -1])
    print("Silhouette Score:", round(score, 3))
else:
    print("Silhouette Score cannot be calculated meaningfully for these clusters.")

# 7. Conclusion
print("\nConclusion:")
print("DBSCAN groups nearby data points into clusters.")
print("Points labeled -1 are considered outliers or noise.")
print("DBSCAN does not require the number of clusters to be specified beforehand.")

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Number of clusters: 3
Number of outliers: 3
Cluster labels: [ 0  1  2  2  2  2  1  2  0  2  0  0  0  0  0  2  1  2
0  2  0  0  2  1
  2  1  1  2  1  0  0  0  2  2  1  1  2  1  2  1  0  1  2  0  2  0  0
2
  0  1  2  1  2  1  1  1  2  0  1  0  2  1  2  2  2  2  1  1  2  1  1
2
  1  0  1  0  0  1  0  1  1  0  0  2  2  2  2  1  0  2  0  1  1  1  0
2
  1  0  0  1  2  2  1  2  1  0  2  0  0  2  0  0  1  2  1  2  2  1  1
1
  2  0  1  2  2  0  1  1  0  1  0  2  1  2  2  0  0  0  1  0  1  1  2
0
  0  1  0  2  2  0  1  2  0  2  0  2  2  1  0  0  1  0  0  2  1  0  0
1
  2  0  2  1  0  2  1  0  1  1  1  0  1  0  2  2  2  1  0  0  0  1  1
0
  2  2  1  2  1  1  0  0 -1 -1 -1]
```




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Silhouette Score: 0.908

Conclusion:
DBSCAN groups nearby data points into clusters.
Points labeled -1 are considered outliers or noise.
DBSCAN does not require the number of clusters to be specified
beforehand.
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